LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Quick Revision: Control and Coordination

Nervous system

- **Neuron:** dendrite (receives information) → cell body → axon (carries the electrical impulse) → nerve ending.
- **Synapse:** the gap between neurons. At the axon end, chemicals are released that cross the synapse and start an impulse in the next neuron.
- **Reflex action:** a quick, automatic response (hand pulled from a hot object, blinking, sneezing). It is controlled by the spinal cord.
- **Reflex arc:** receptor → sensory neuron → relay neuron in the spinal cord → motor neuron → effector (muscle or gland).

Part of brain	Main functions
Forebrain (cerebrum)	Thinking, intelligence, memory, voluntary actions, sensory areas (sight, hearing, smell), hunger
Midbrain	Involuntary actions, e.g. reflex movements of the head and eyes
Hindbrain: cerebellum	Posture and balance; precision of voluntary actions (walking in a straight line, picking up a pencil)
Hindbrain: medulla	Blood pressure, salivation, vomiting
Hindbrain: pons	Involuntary actions, e.g. regulation of breathing

Protection: the brain sits inside the skull in a fluid-filled balloon (shock absorber). The spinal cord is protected by the vertebral column.

Coordination in plants

- **Touch-me-not (Mimosa):** leaves fold when touched. Cells change shape by changing their water content. No growth is involved.
- **Tropisms** are growth movements towards or away from a stimulus. Phototropism (light): shoot +, root –. Geotropism (gravity): root +, shoot –. Hydrotropism (water) and chemotropism (pollen tube growing towards the ovule).

Plant hormone	Effect
Auxin	Made at the shoot tip; makes cells grow longer; causes bending towards light
Gibberellin	Growth of the stem
Cytokinin	Promotes cell division (high in fruits and seeds)
Abscisic acid	Inhibits growth; causes wilting of leaves

Hormones in animals

Gland	Hormone	Function / problem
Pituitary	Growth hormone	Growth. Deficiency in childhood → dwarfism
Thyroid	Thyroxin	Metabolism of carbohydrates, fats, proteins. Needs iodine (lack → goitre)
Pancreas	Insulin	Controls blood sugar. Deficiency → diabetes
Adrenal	Adrenaline	Emergencies: faster heartbeat and breathing, blood diverted to muscles
Testes	Testosterone	Changes in boys at puberty
Ovaries	Oestrogen	Changes in girls at puberty

Hypothalamus	Releasing hormones	Stimulates the pituitary (e.g. growth hormone release)

Feedback mechanism: controls the timing and amount of hormone. If blood sugar rises, the pancreas releases more insulin. As sugar falls, insulin secretion is reduced.

Quick check

1. Which part of the brain maintains posture and balance?

2. Why is iodised salt advised in our diet?

3. How is the movement of a touch-me-not leaf different from a shoot bending towards light?